

The most widely cited theories are behavioral. In the main behavioral theories, momentum arises because of the biased way that investors interpret or act on information. Suppose good news on a stock comes out. Momentum can be generated in two ways. First, investors could have delayed overreaction to this news, causing the price to persistently drift upward. Second, investors could underreact to the news. The price initially goes up, but it does not go up as much as it should have to fully reflect how good the news actually was. Investors then learn and cause the stock to go up again the next period. Behavioral explanations, then, fall into two camps: momentum is an overreaction phenomenon, or it is an underreaction phenomenon. Distinguishing between these camps is difficult and still bedevils the literature.³⁶

The seminal overreaction models are Barberis, Shleifer, and Vishny (1998) and Daniel, Hirshleifer, and Subrahmanyam (1998). Barberis, Shleifer, and Vishny's investors suffer from *conservatism bias*, which causes them to overreact to information because they stick doggedly to their prior beliefs. This causes momentum. In the Daniel, Hirshleifer, and Subrahmanyam model, investors also have psychological biases giving rise to momentum. In this model, investors are *overconfident* and overestimate their abilities to forecast firms' future cash flows. They also have biased *self-attribution*: when they are successful, it must be due to their skill, and when they are unsuccessful, it must be due to bad luck. Informed, overconfident investors (think of retail investors and overconfident hedge fund managers) observe positive signals about some stocks that perform well. These overconfident investors attribute the good performance to their own skill, leading to overconfidence. Based on increased confidence, they overreact and push up the prices of stocks above their fundamental values, generating momentum.

The standard reference for the underreaction theory is Hong and Stein (2000). Hong and Stein rely on "bounded rational" investors who have limited information. Momentum in Hong and Stein's model is caused by "news watchers" who receive signals of firm value but ignore information in the history of prices. Other investors trade only on past price signals and ignore fundamental information. The information received by the news watchers is received with delay and is only partially incorporated into prices when first revealed to the market. This causes underreaction.

In both the underreaction and overreaction models, prices eventually reverse when they revert to fundamentals in the long run.

Implications for Asset Owners

In the context of these behavioral stories, the asset owner should think about what types of psychological biases that she has and how these biases differ from those of the average investor. Do you overreact (or underreact) in a way similar

³⁶ See, for example, Jegadeesh and Titman (2001).